

Homework 2

Due Wednesday, 9 September 2026

1. The Weibull density function is given by

$$f(x; \alpha, m) = \begin{cases} \frac{1}{\alpha} m x^{m-1} e^{-x^m/\alpha} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where α and m are positive constants. The Weibull density function is often used as a model for the lifetimes of physical systems.

- a. Let Y follow an exponential distribution with mean parameter β . Prove that $W = \sqrt{Y}$ has a Weibull distribution with $\alpha = \beta$ and $m = 2$. (Hint: this is a transformation problem)
 - b. If a random sample of size n ($X_1, \dots, X_n$) is taken from a Weibull distribution, find the pdf of the smallest order statistic $X_{(1)}$. What distribution, with parameters, does $X_{(1)}$ follow?
2. Let $X_1, \dots, X_n$ be a random sample from the Beta($\alpha = 2, \beta = 2$) distribution. Find the pdf of the largest order statistic, $X_{(n)}$.
 3. Find a method-of-moments estimator (MME) of θ based on a random sample $X_1, \dots, X_n$ from each of the following distributions.
 - a. $f(x; \theta) = \theta(1 - \theta)^{x-1}, x = 1, 2, \dots; 0 \leq \theta \leq 1$
 - b. $f(x; \theta) = (\theta + 1)x^{-\theta-2}, x > 1; \theta > 0$
 - c. $f(x; \theta) = \theta^2 x e^{-\theta x}, x > 0; \theta > 0$
 4. Let $X_1, \dots, X_n$ be a random sample from the pdf

$$f(x; \theta) = \theta x^{\theta-1}, \quad 0 \leq x \leq 1; 0 < \theta < \infty$$

Find a method-of-moments estimator of θ .